CS 31204: Computer Networks – Moving From End-to-End To Per Hop

Department of Computer Science and Engineering

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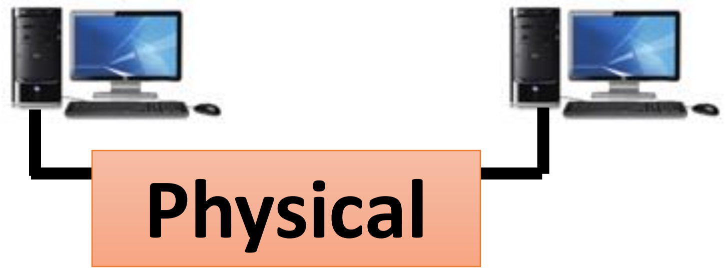
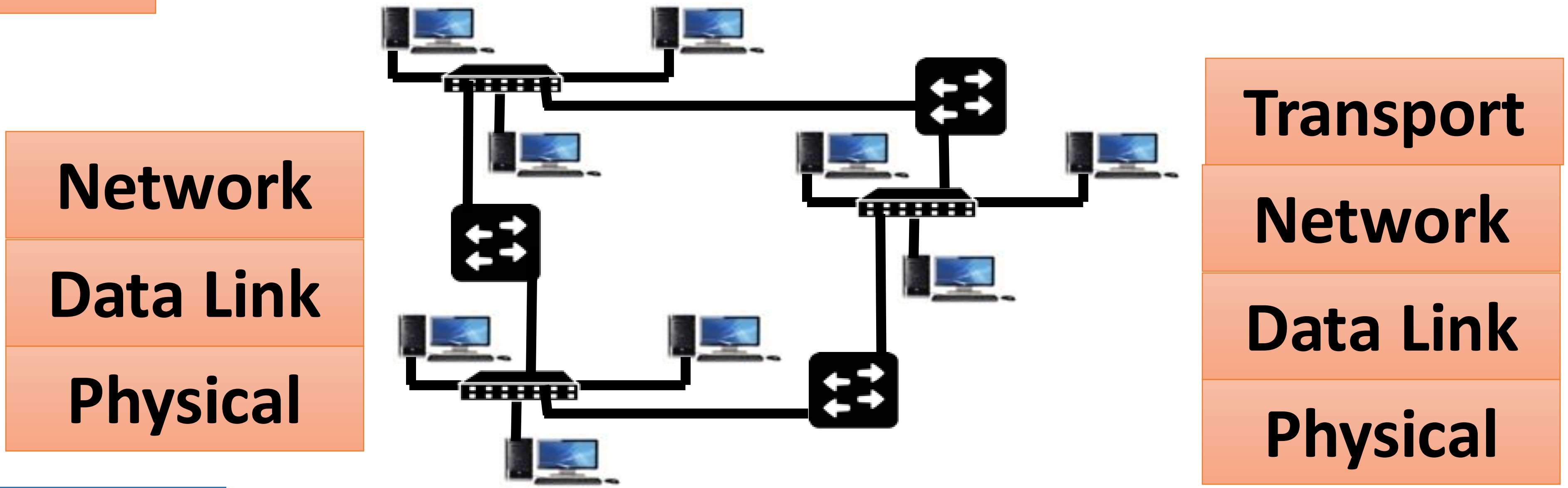
What We Have Learnt So Far …

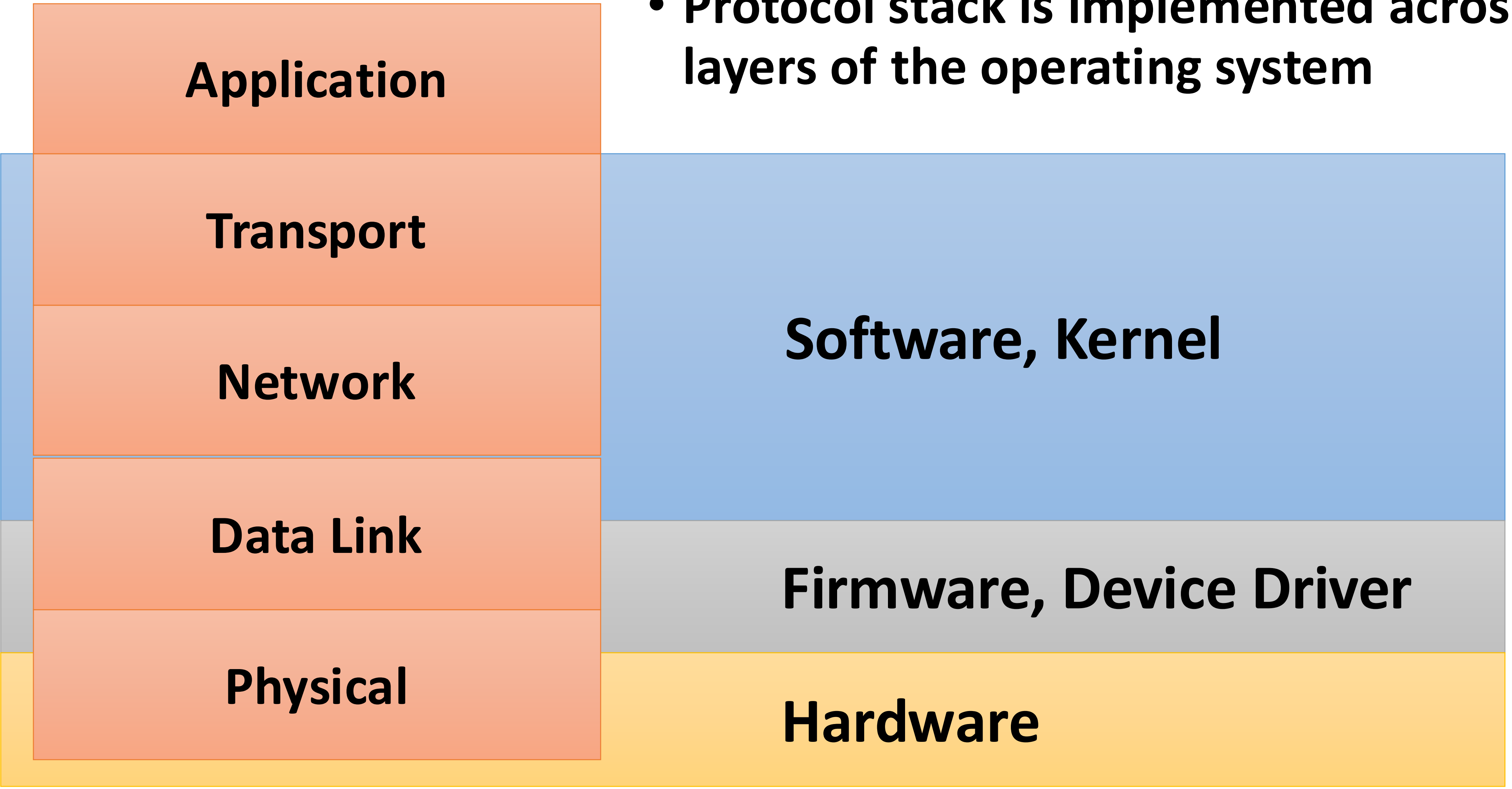
- The design of current network architecture is based on experience and requirements



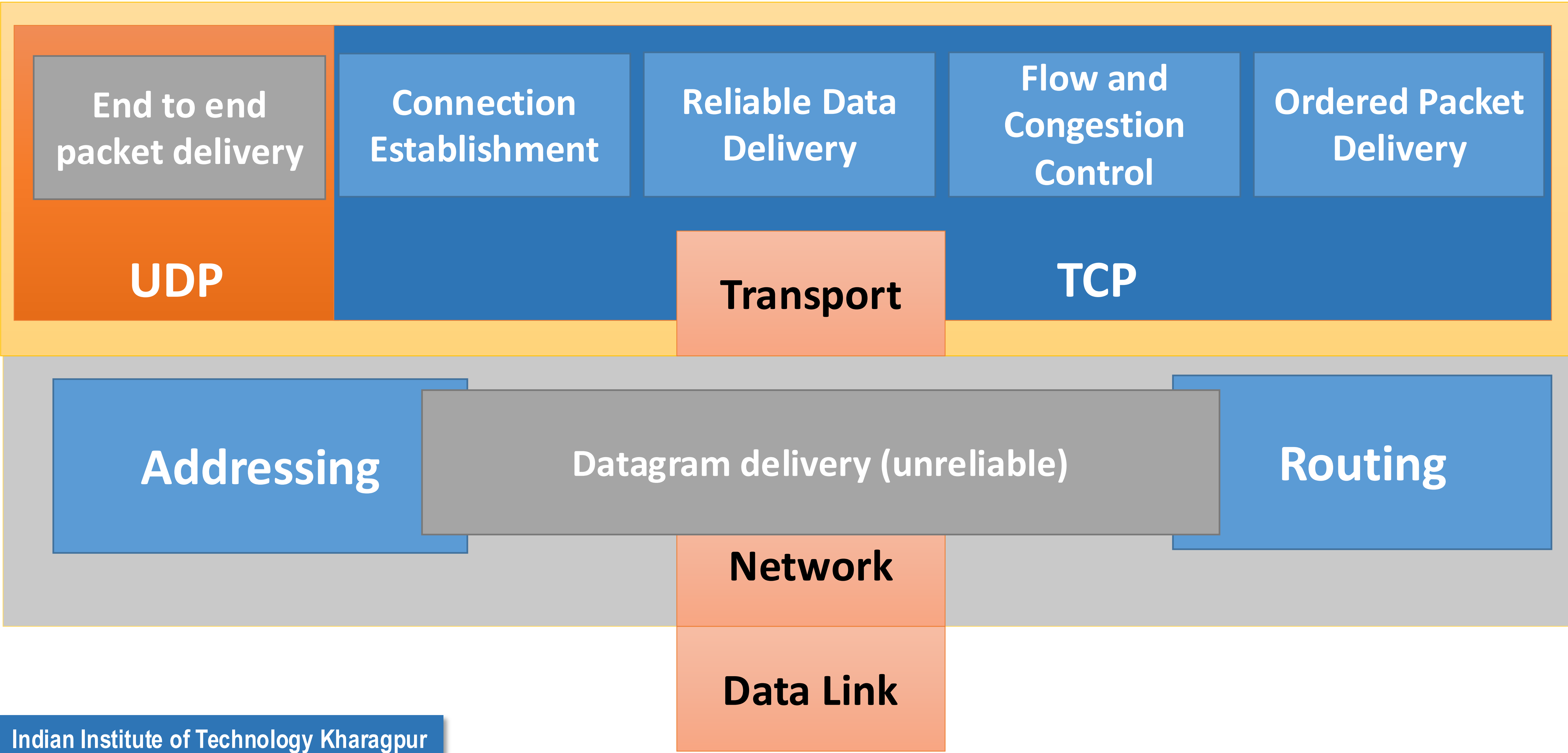


What We Have Learnt So Far …

- Protocol stack is implemented across different layers of the operating system

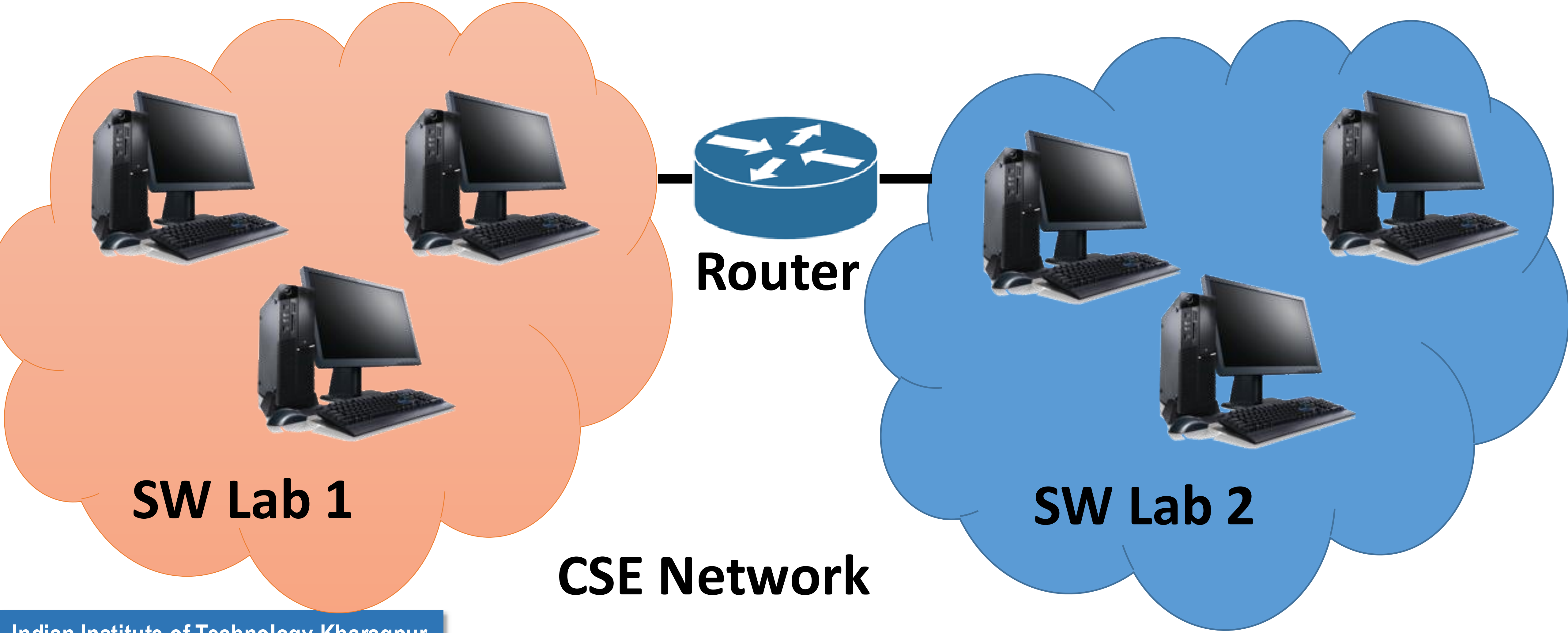


Network (Internet) Layer Services



Internet Architecture – Basic Principles

- Internet is organized in a hierarchical fashion.



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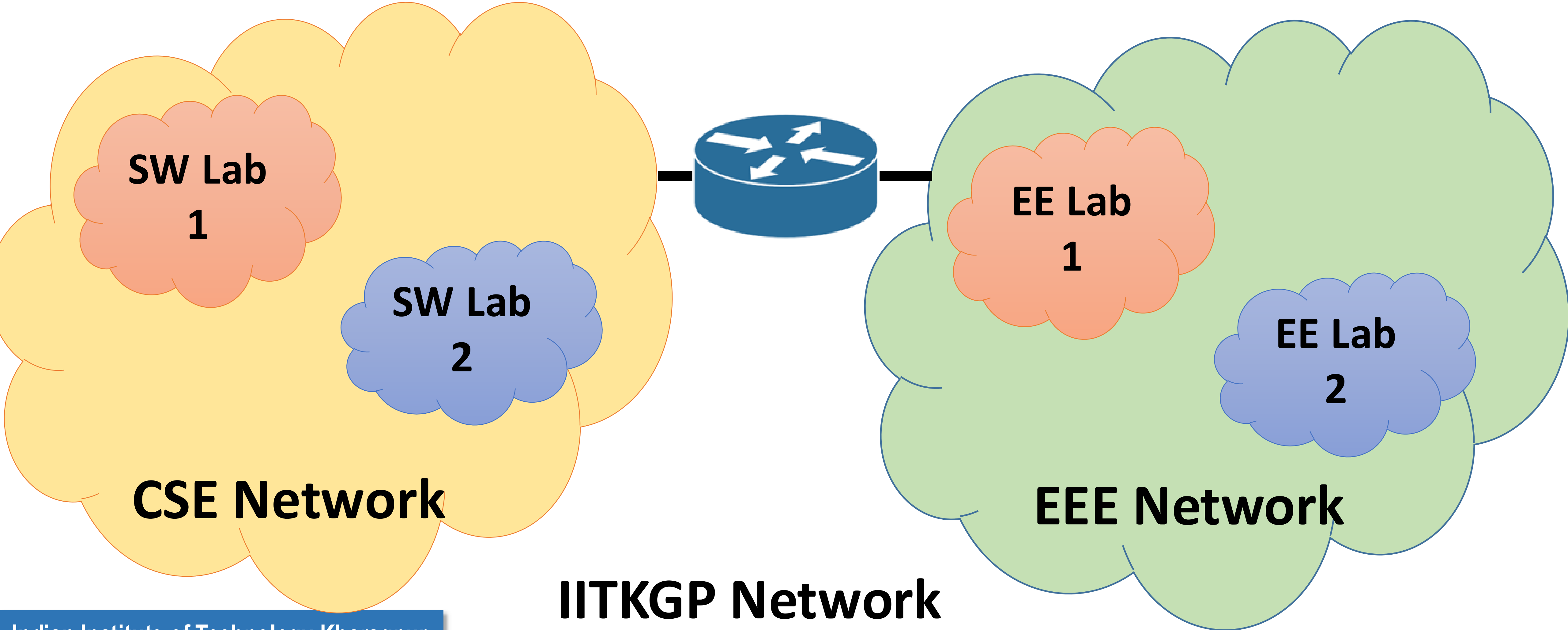

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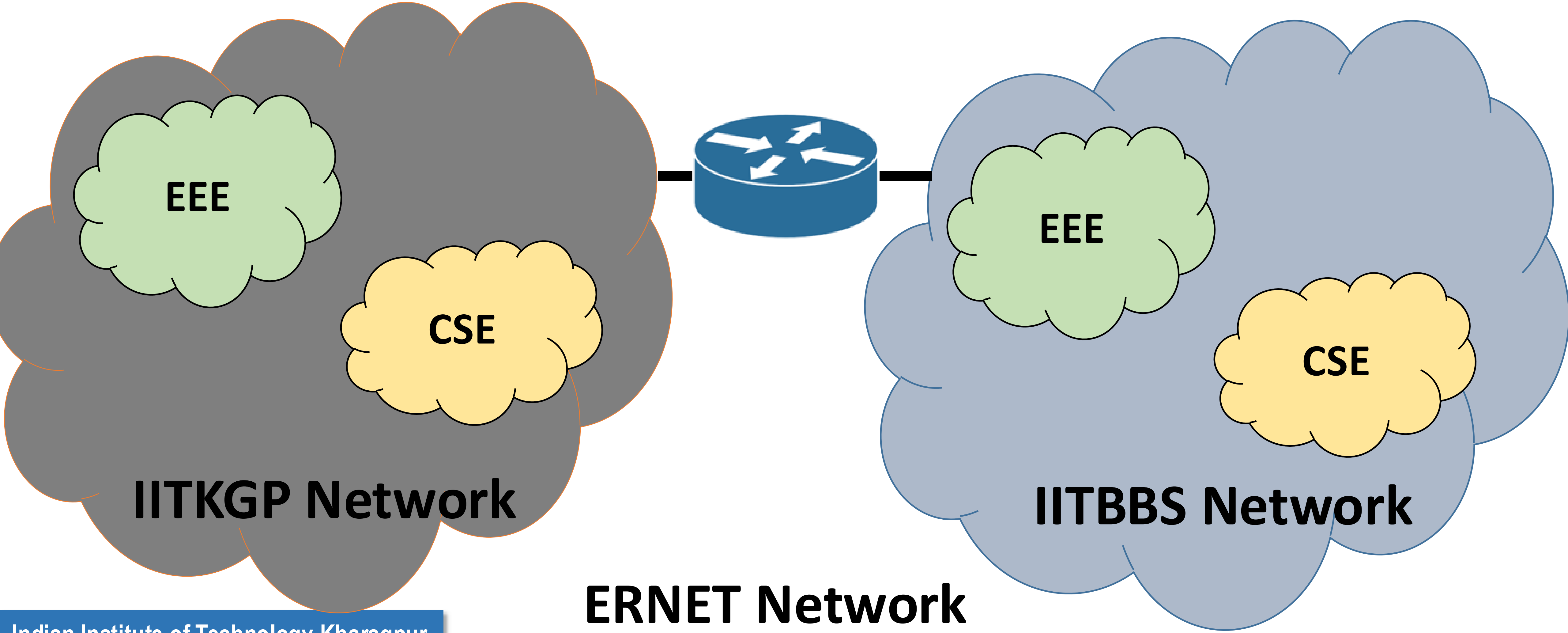

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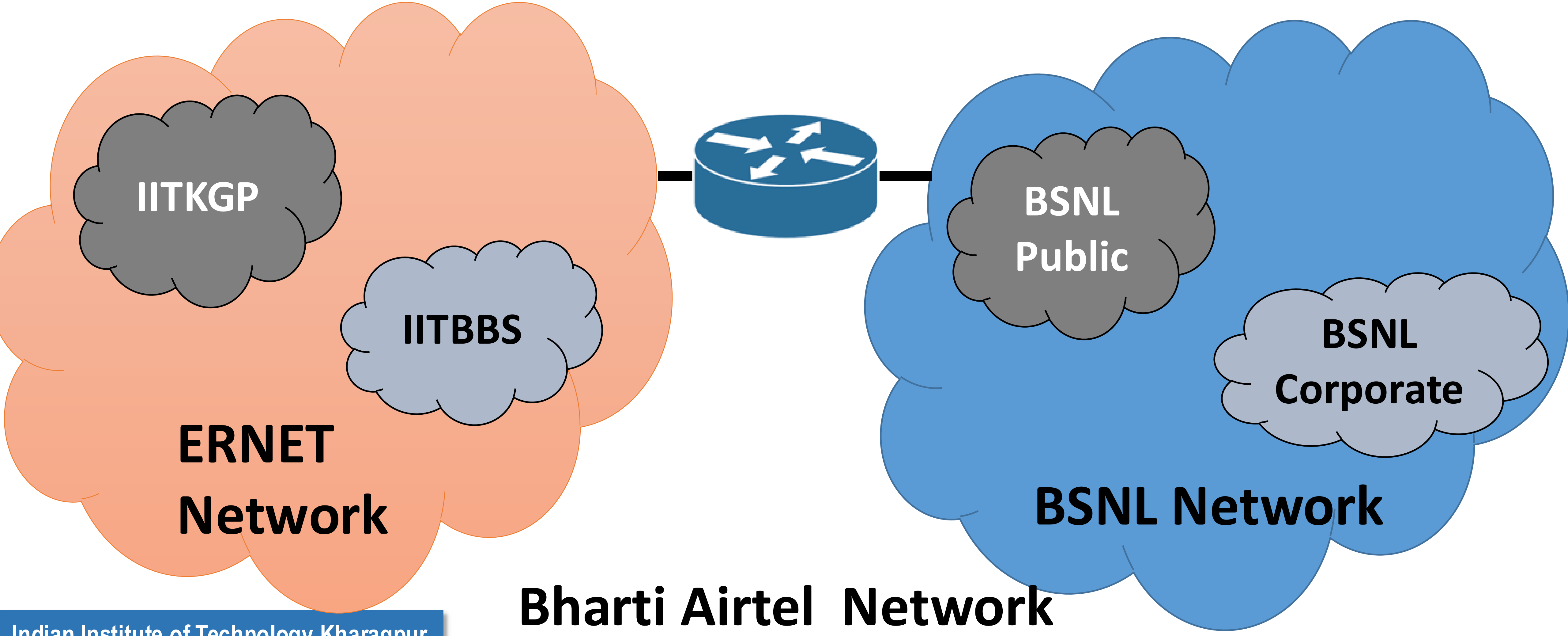
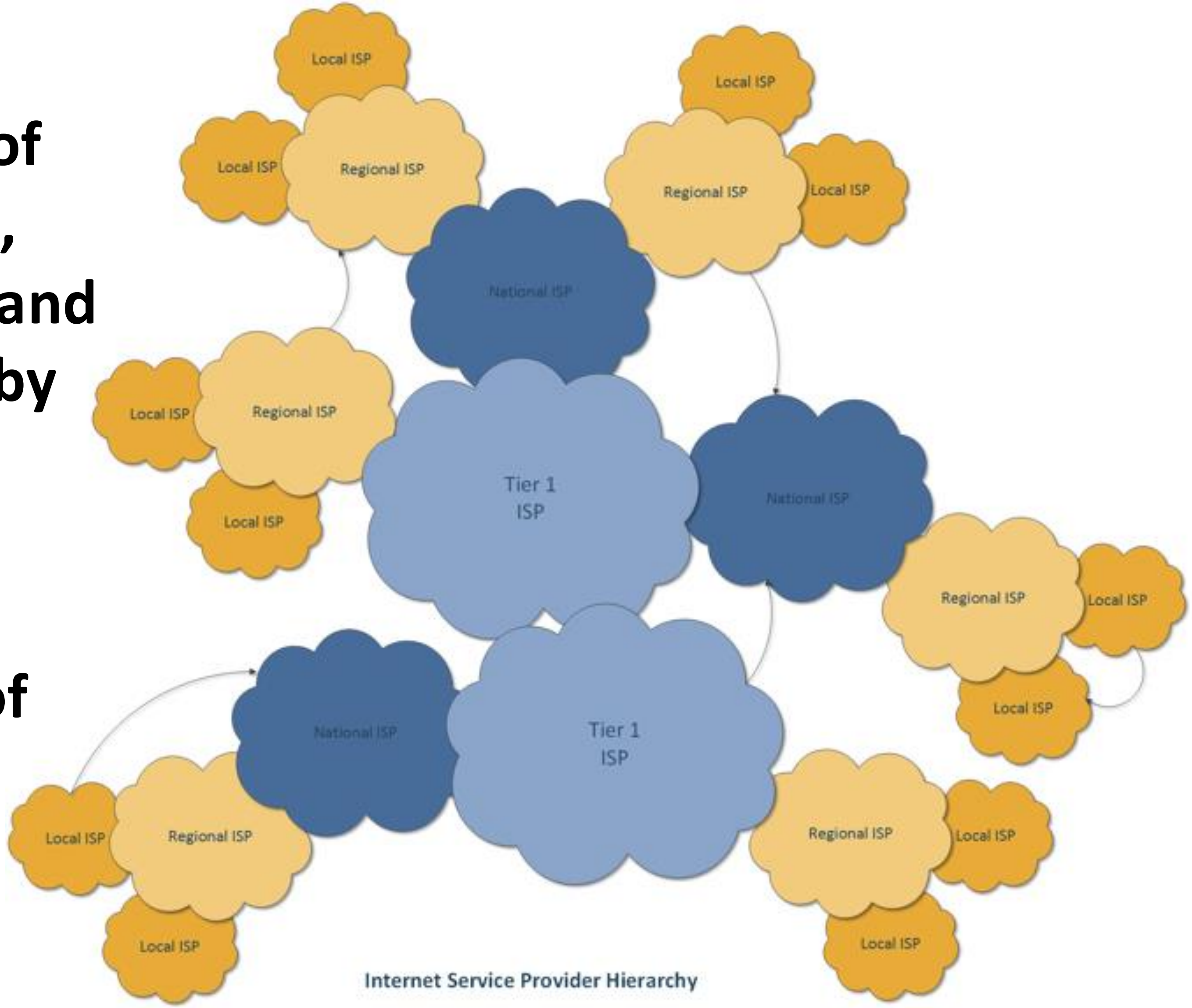


Internet Architecture

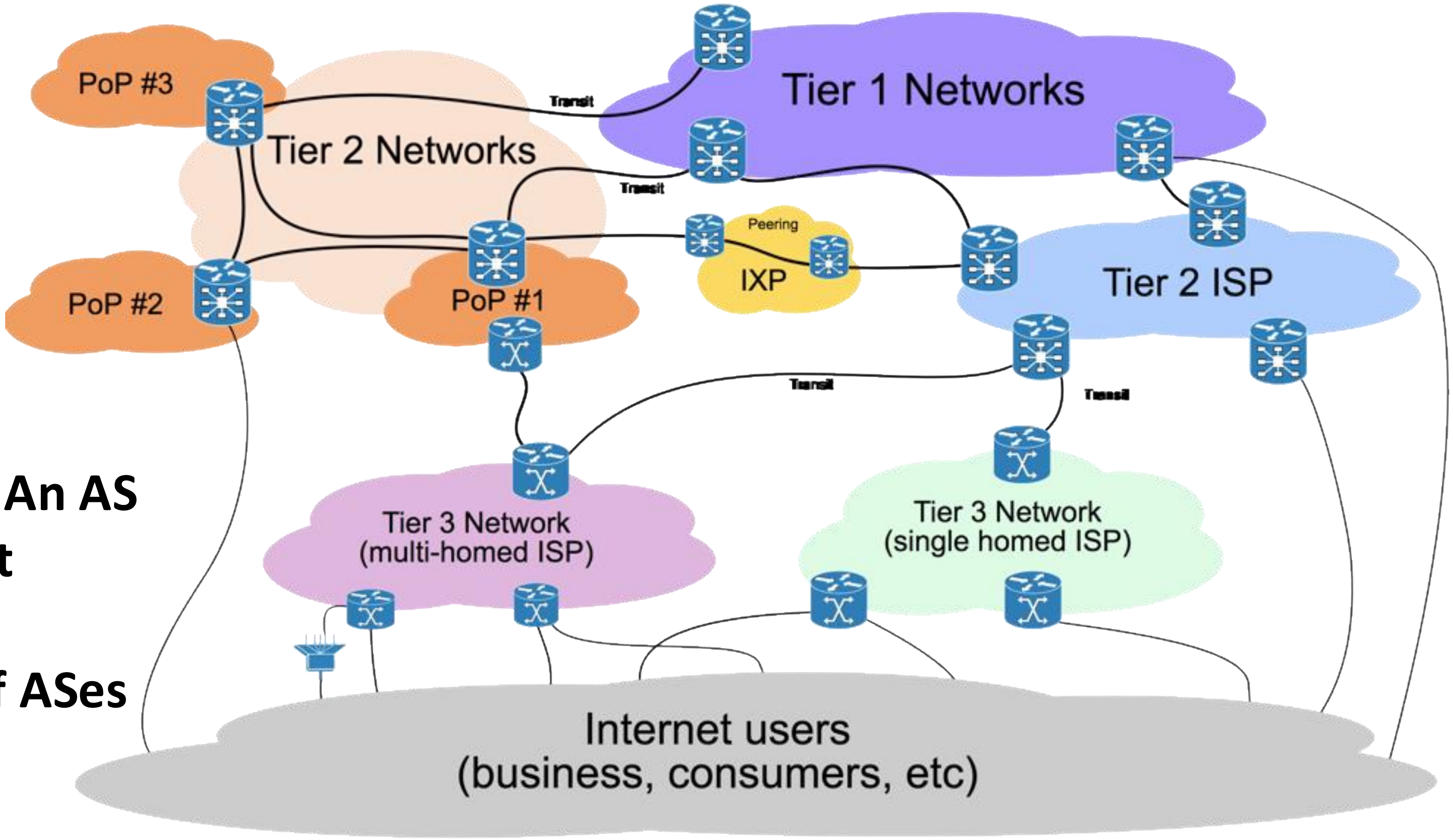
Autonomous Systems (AS) – A set of LANs for an administrative domain, identified by a unique AS number, and the routing policies are controlled by a single administrator.

Local Area Network (LAN) – A set of devices with a common layer 3 gateway



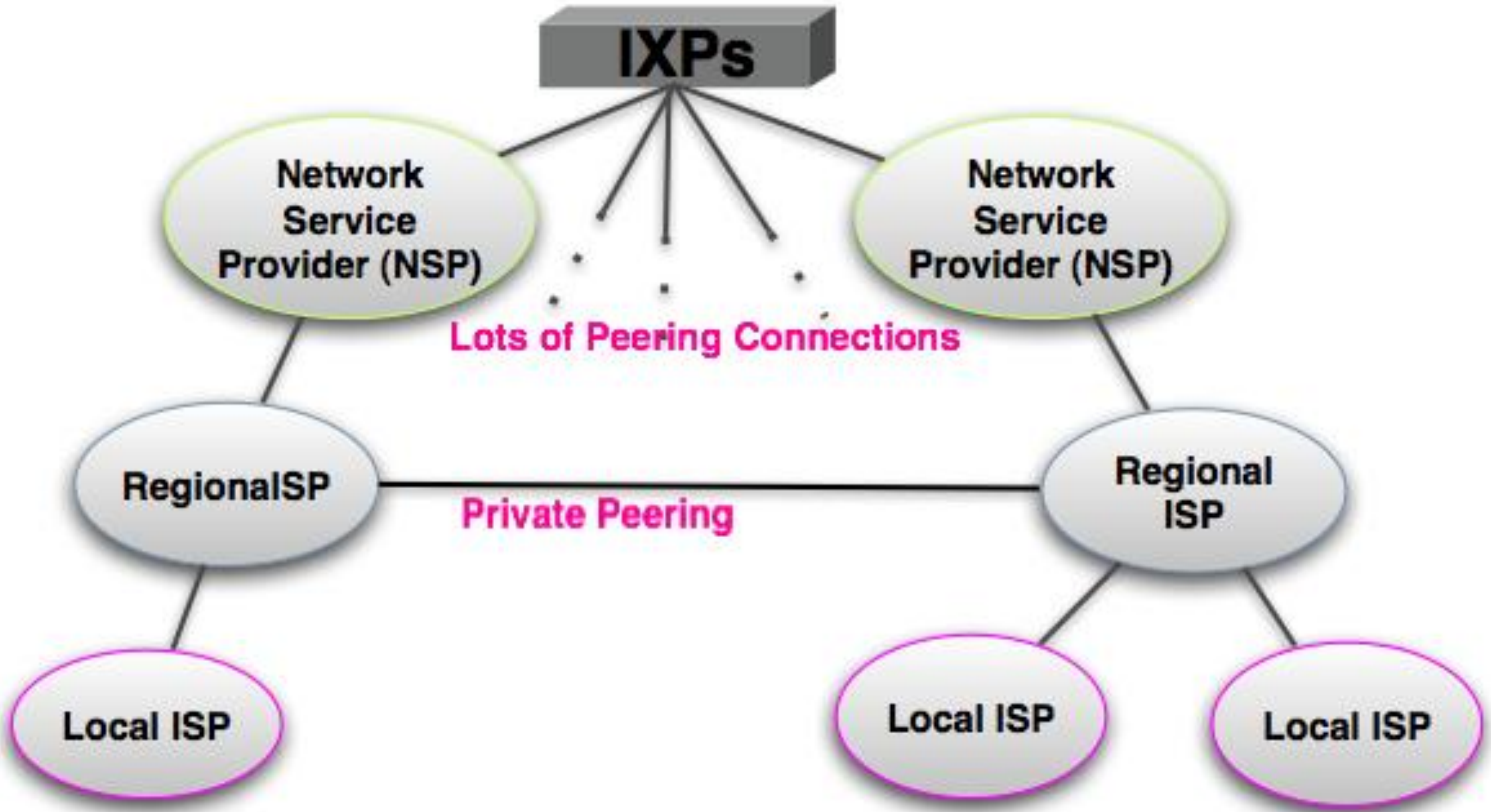
Internet Service Provider Hierarchy

Internet Architecture

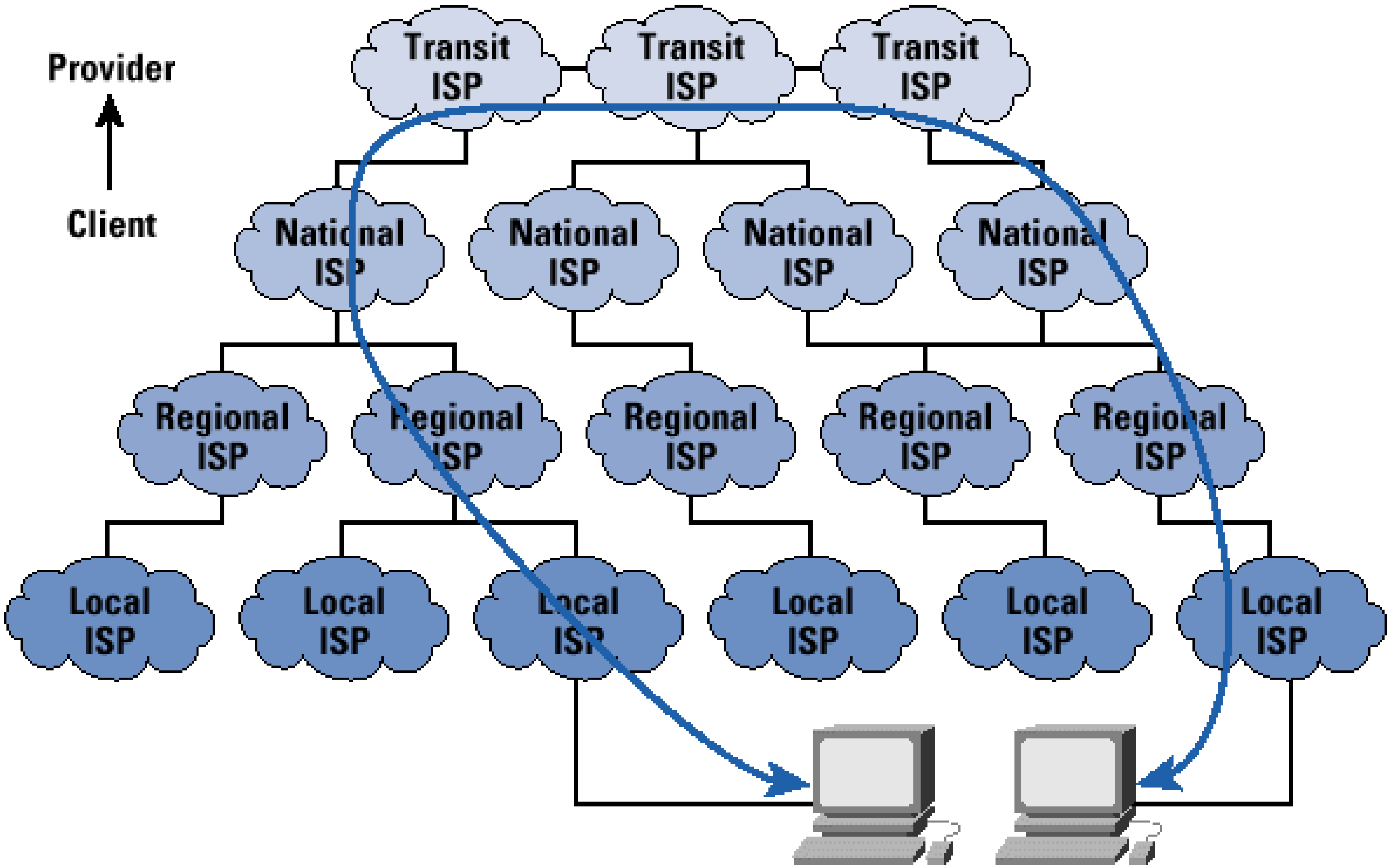


Internet Service Providers (ISP) – An AS provides Internet connectivity to another group of ASes or end users

Peering between ISPs

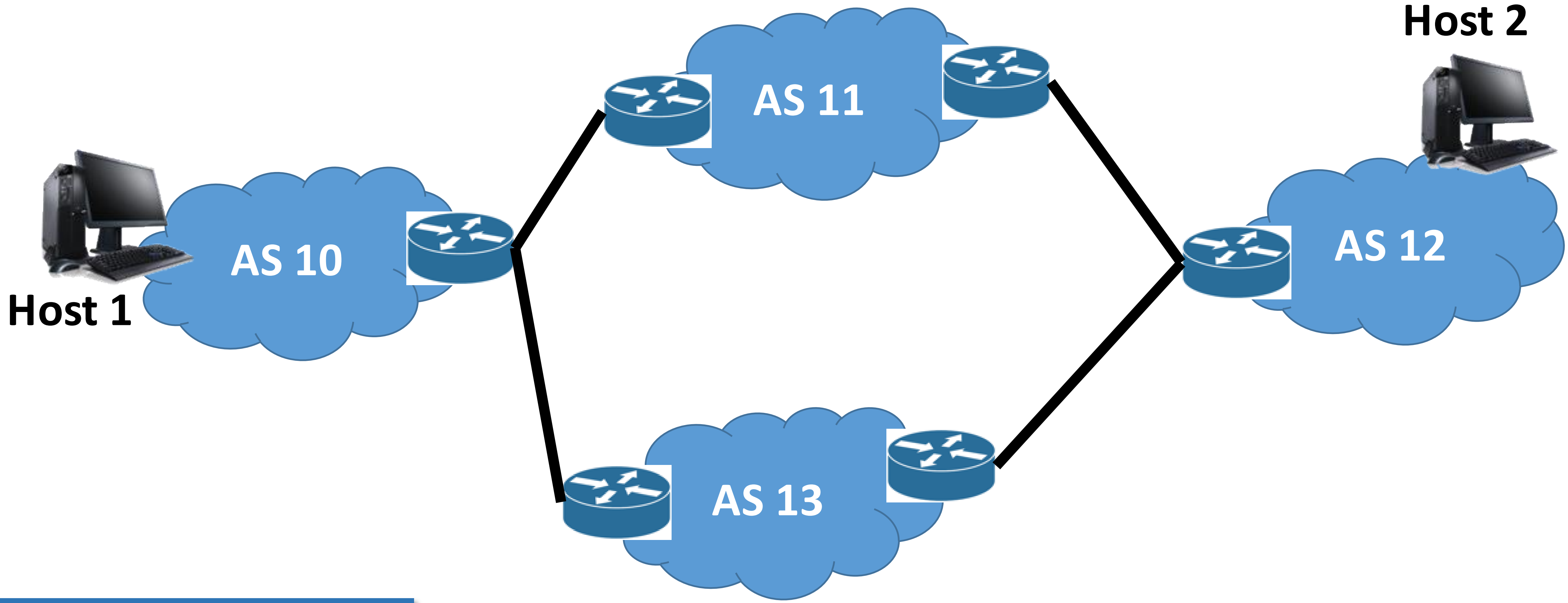


Communication between Two Nodes over ISPs



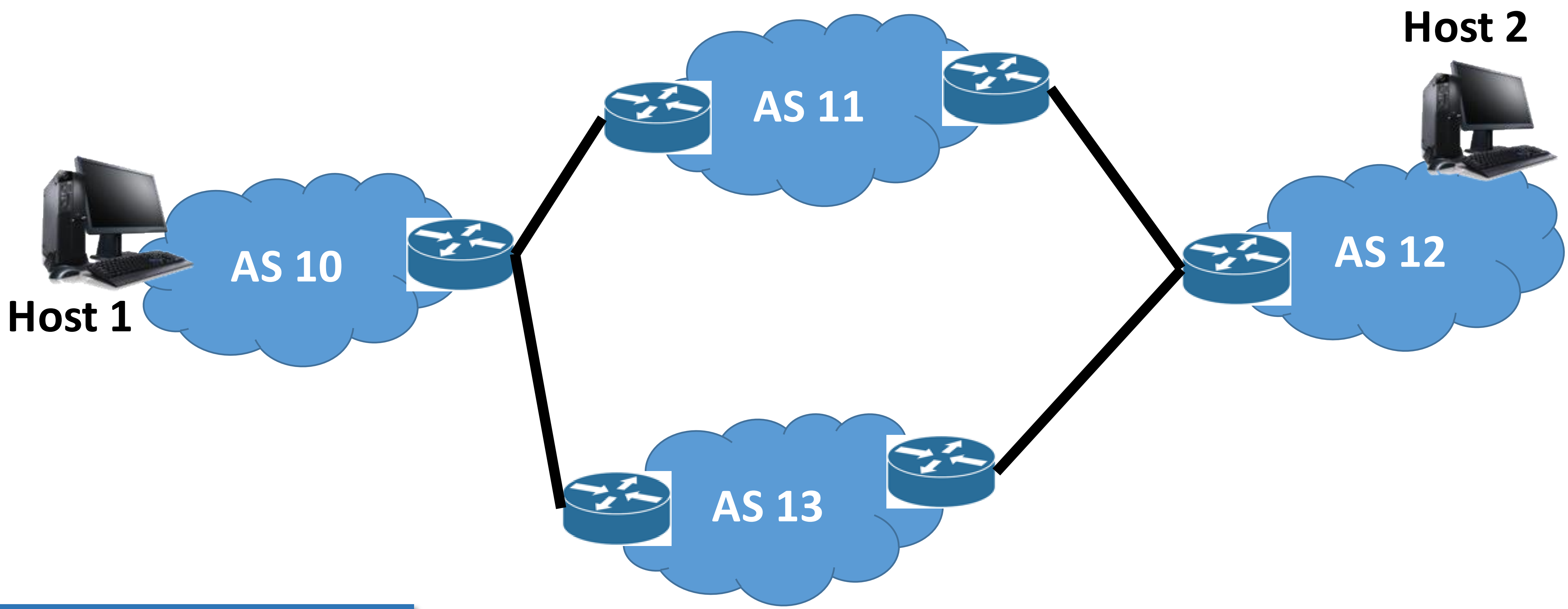
IP Addressing – Basic Principles

- We need to forward data packets from one network to another network via different intermediate networks.



IP Addressing – Basic Principles

- The address should identify a network as well as a host inside a network



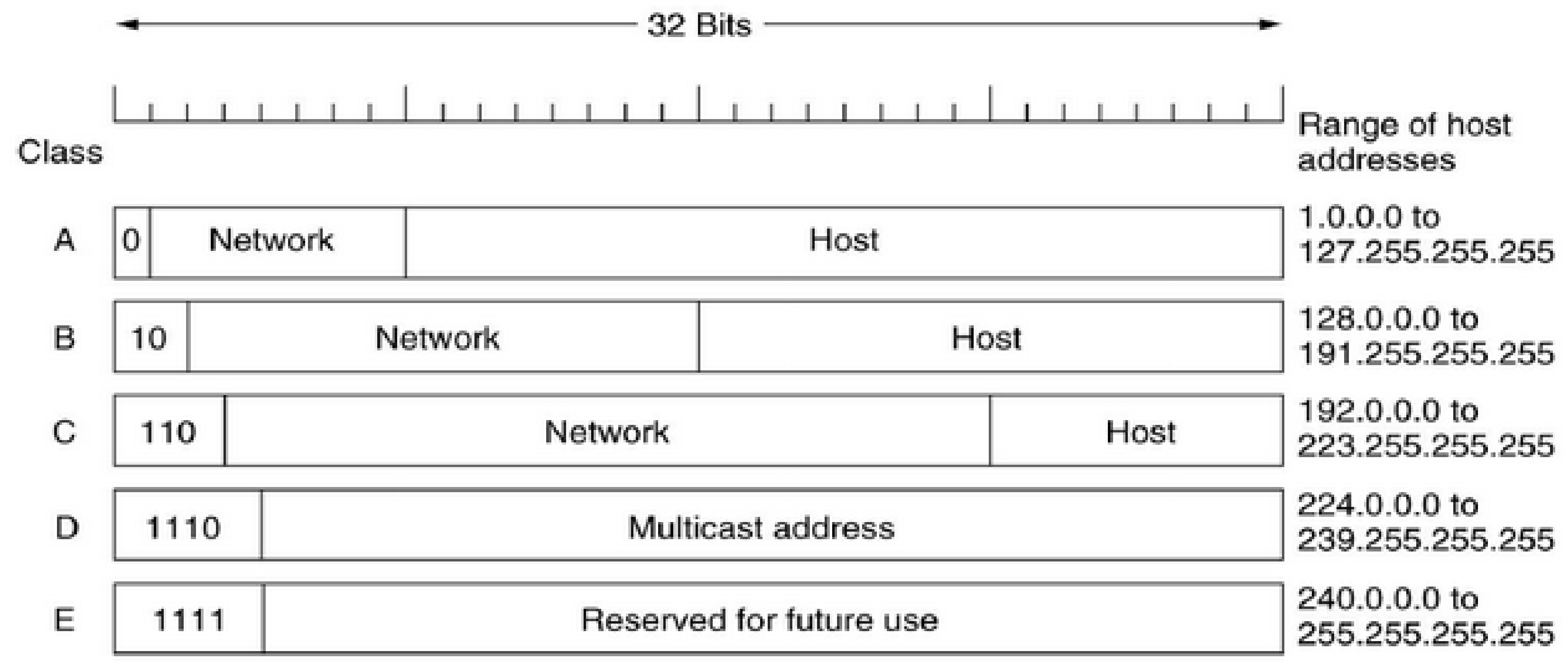
IP Addressing

Network address	Host address

- Divide the address space (32 bit in IPv4) among network address and host address

- **The old age – Classful addressing:** Fixed number of bits for network address and host address

Classful Addressing



- **How to identify a class –** use the first few bits
    - **0 – Class A, 10 – Class B, 110 – Class C, 1110 – Class D, 1111 – Class E**

Network Address and Broadcast Address

- **Network address –** identify a network
  - All 0's in the host address part
  - **Ex-1 (Class A):** 01111110.00000000.00000000.00000000 (126.0.0.0)
  - **Ex-2 (Class B):** 10111101.11101001.00000000.00000000 (189.233.0.0)

- **Broadcast address –** send the data to **all the hosts** of a network
  - All 1's in the host address part
  - **Ex-1 (Class A)**: 01111110.11111111.11111111.11111111 (126.255.255.255)
  - **Ex-2 (Class B):** 10111101.11101001.11111111.11111111 (189.233.255.255)

- **How many valid hosts can be there in a Class A, in a Class B and in a Class C IP address?**

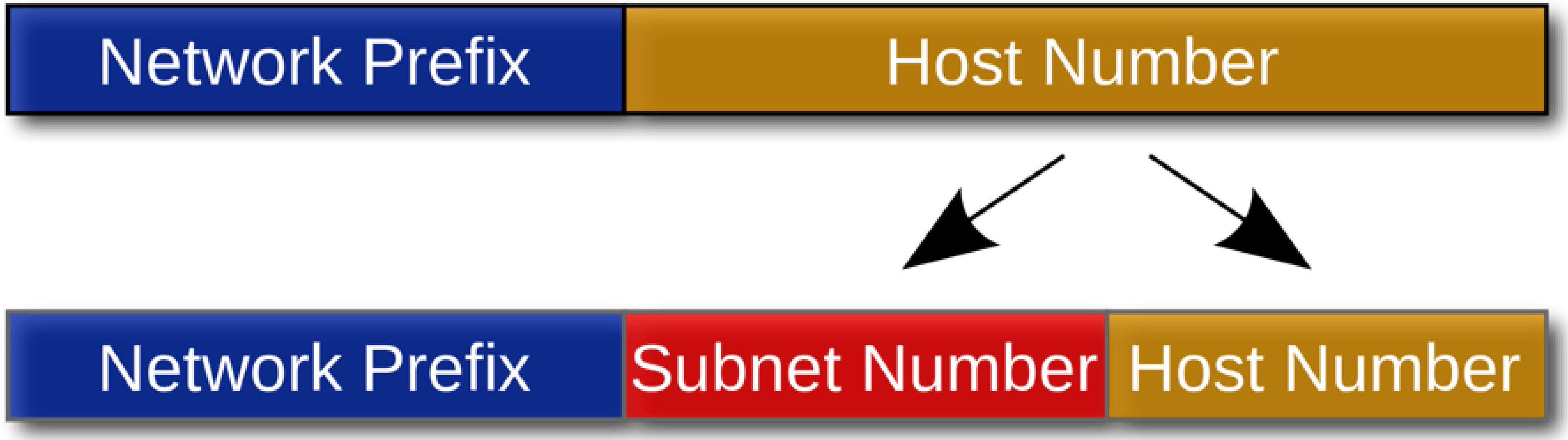
Subnetting and Supernetting – Classless Inter-domain Routing (CIDR)

- **You have 255 hosts in a network. Which IPv4 address class will you use – Class C or Class B ?**
  - **Class C – not possible**
  - **Class B – huge address space is lost (using only 255 addresses out of possible 2^{16}-2 addresses)**

- Split a large network or combine multiple small networks for efficient use of address space
  - **Subnetting** – divide a large network into multiple small networks
  - **Supernetting** – combine multiple small networks into a single large network

- **Subnet mask –** denote the number of bits in the network address field

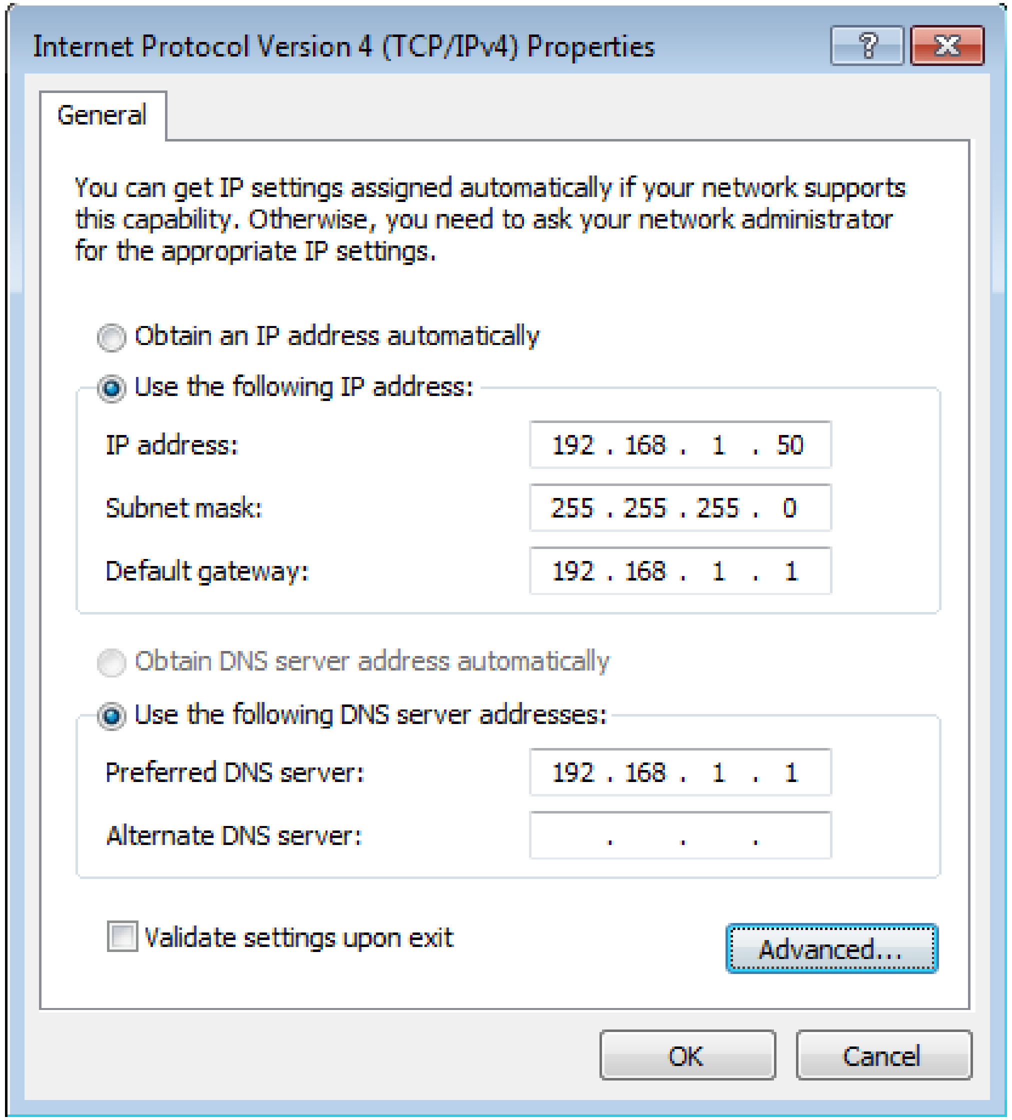
Divide a Network into Subnets

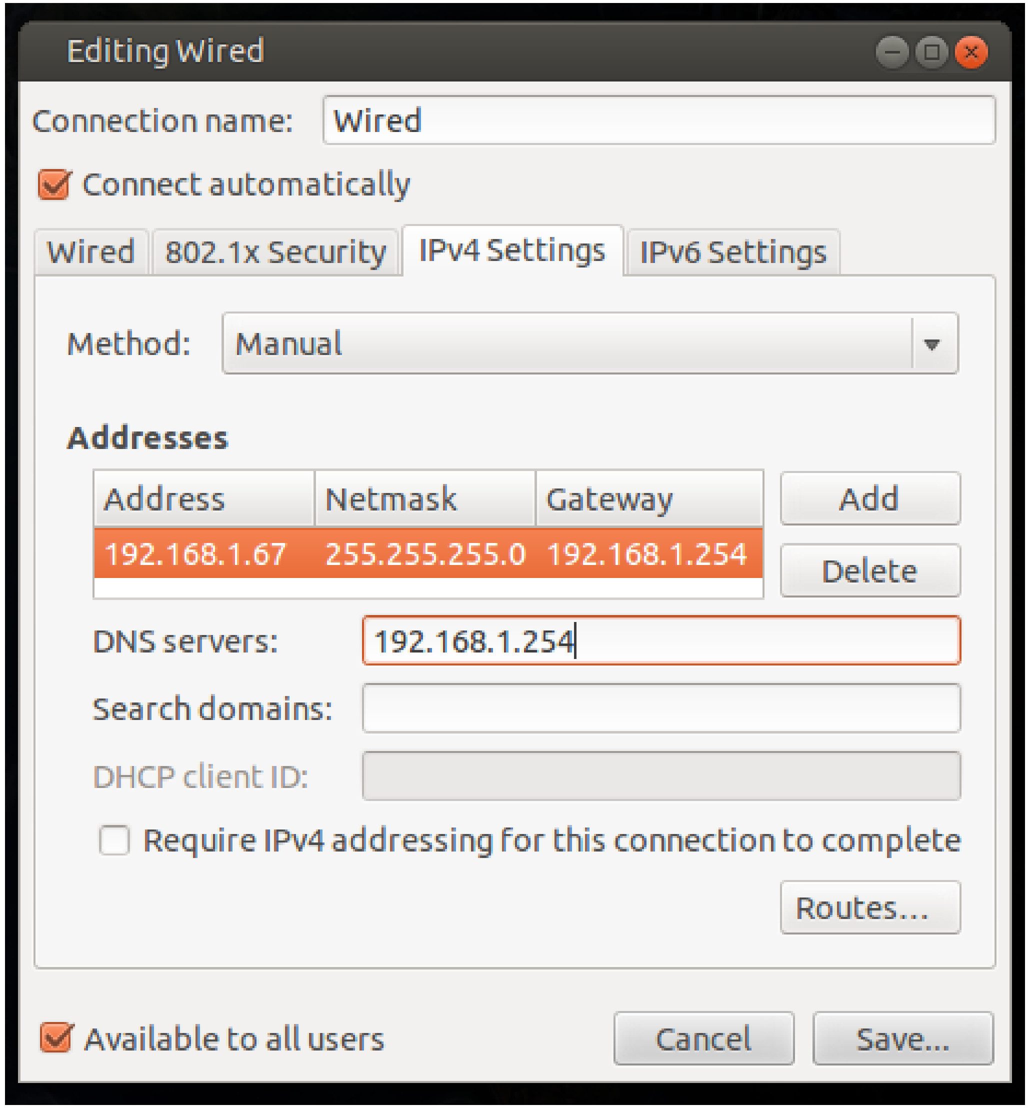


CIDR – Addressing Format

- We write the IP address as 191.180.83.235/12 in CIDR notation
  - The first 12 bits are the network address and rest (32-12)=20 bits are for host address
- The subnet mask is 255.240.0.0

CIDR - Manual IP Setting in the OS

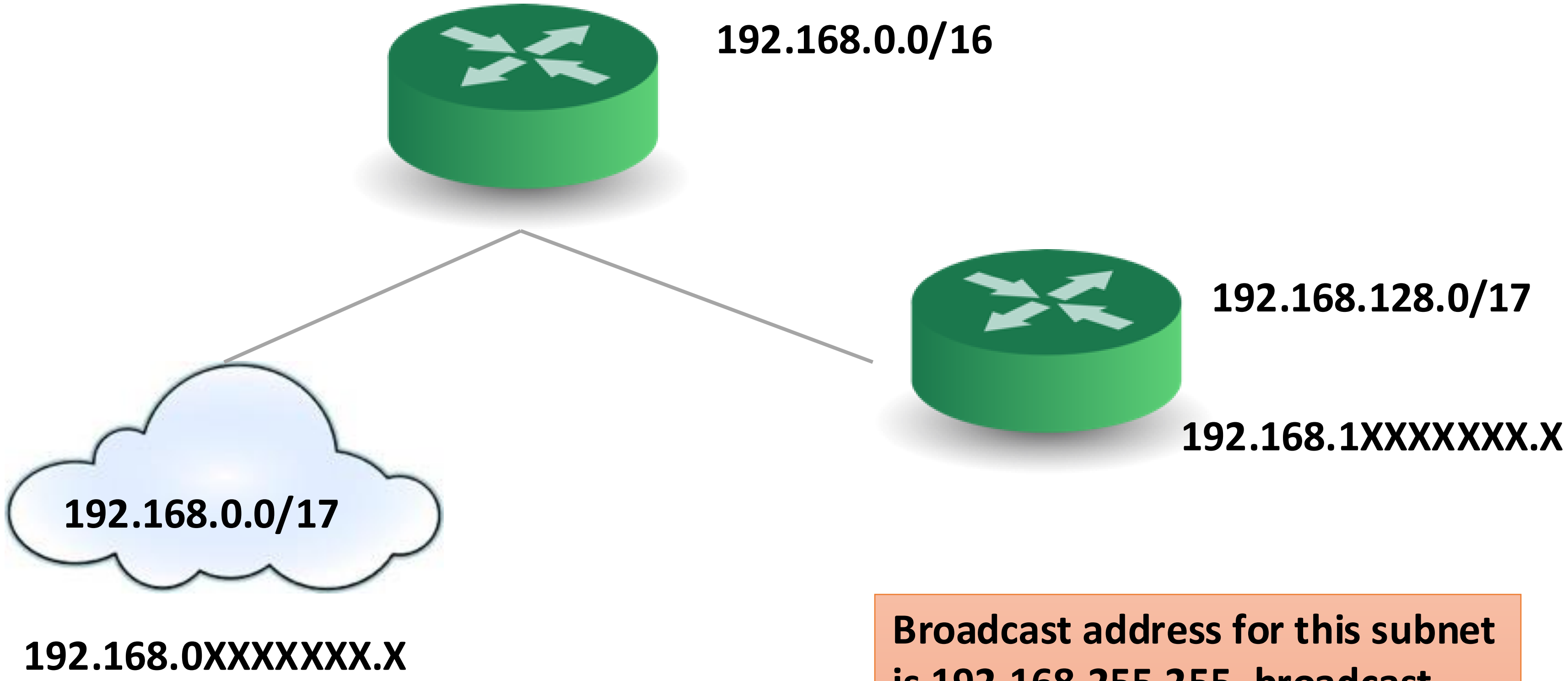




Divide a Network into Subnets

- Let the IP address of a network is 203.110.0.0/16
- We want to divide this network into three subnets
- We need 3 bits for subnets – **why not 2 bits?**
  - Subnet 1 – 100, Subnet 2– 101, Subnet 3 – 110
- Rest 13 bits are used for addressing the hosts of those subnets.
- The subnets are – 203.110.128.0/19, 203.110.160.0/19, 203.110.192.0/19

All Zero and All One Subnets

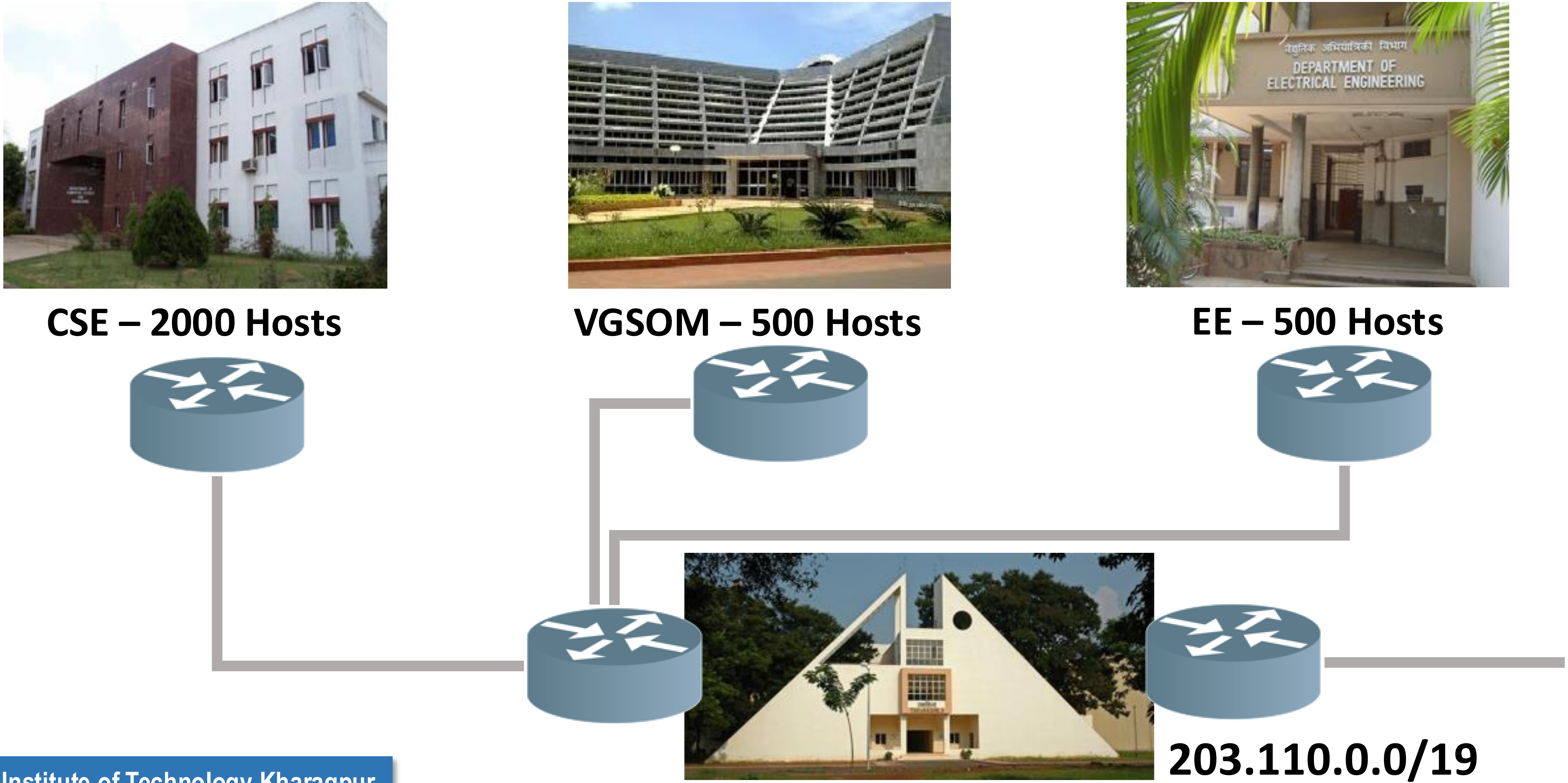


The network address for the subnet and the original network is identical – **Subnet Zero**

Broadcast address for this subnet is 192.168.255.255, broadcast address for the original network is also 192.168.255.255 – **All-One Subnet**

We normally avoid "all zero" and "all one" subnets.

CIDR Example



CIDR Example

CSE – 2000 Hosts

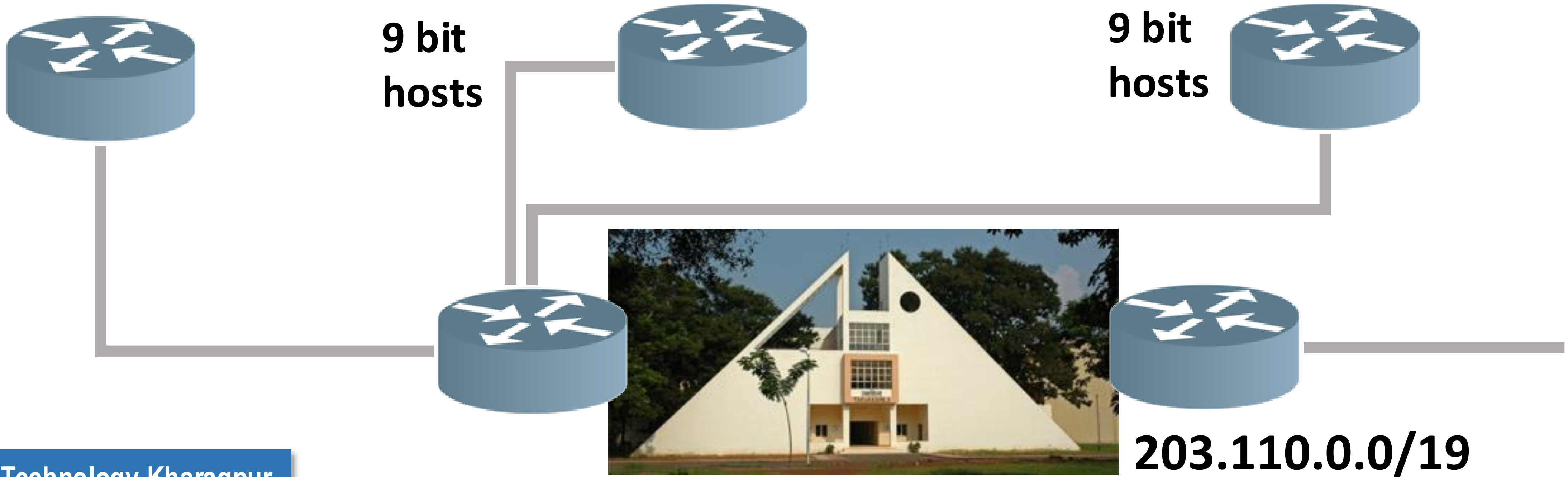
VGSOM – 500 Hosts

EE – 500 Hosts

11 bit
hosts

9 bit
hosts

9 bit
hosts

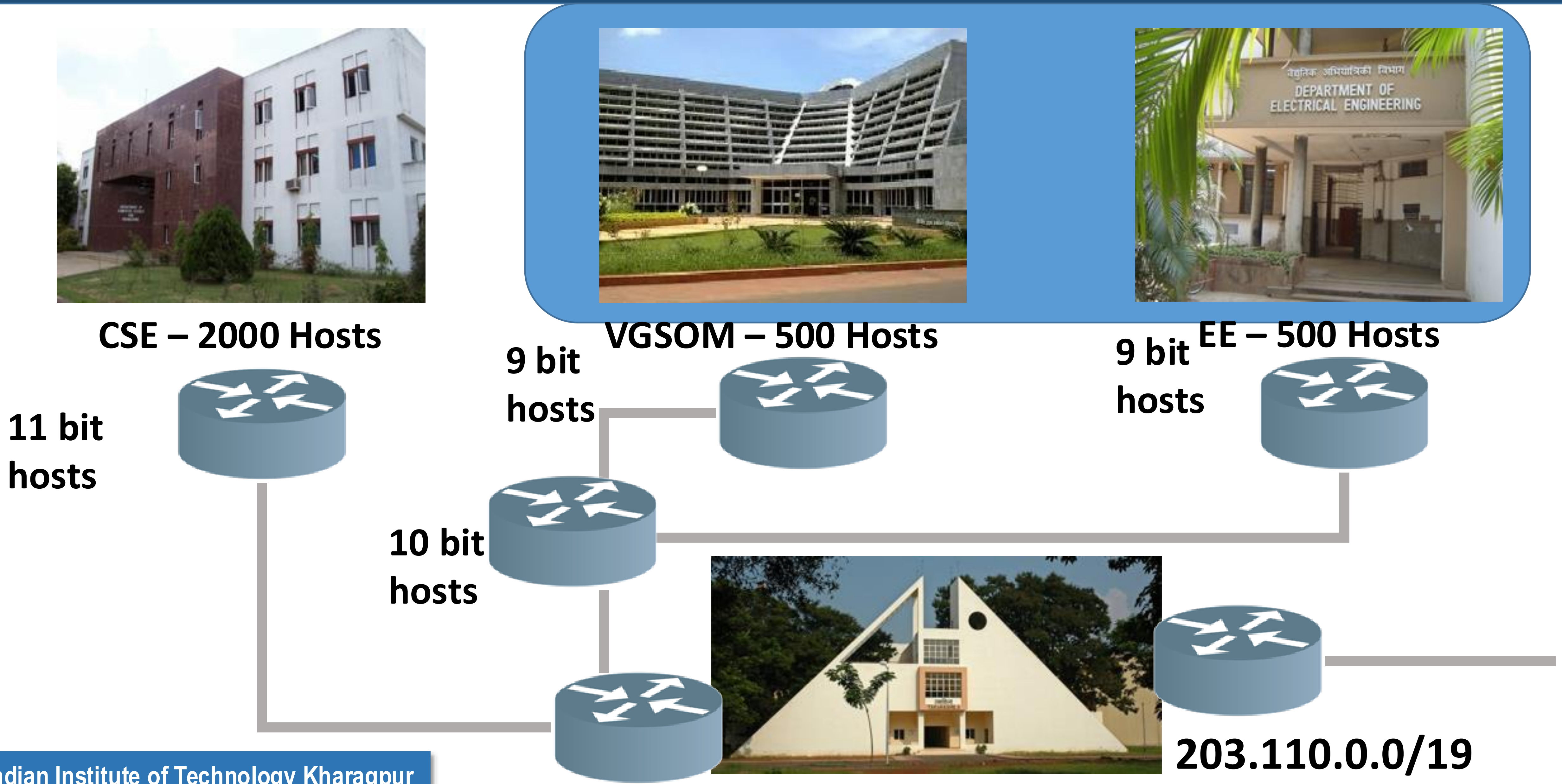


203.110.0.0/19

CIDR Example

- Address space – 203.110.0.0/19
  - 13 bits are available to serve all the hosts of IITKGP network
  - We need to divide these address space among 3 subnets
- CSE – 11 bits, VGSOM – 9 bits, EE – 9 bits for host address
- We have 2 bits left for identifying three subnets **– Is this possible?**
  - Avoid “all zero” and ”all one” subnets
- Let us apply CIDR – Combine VGSOM and EE Networks together

CIDR Example

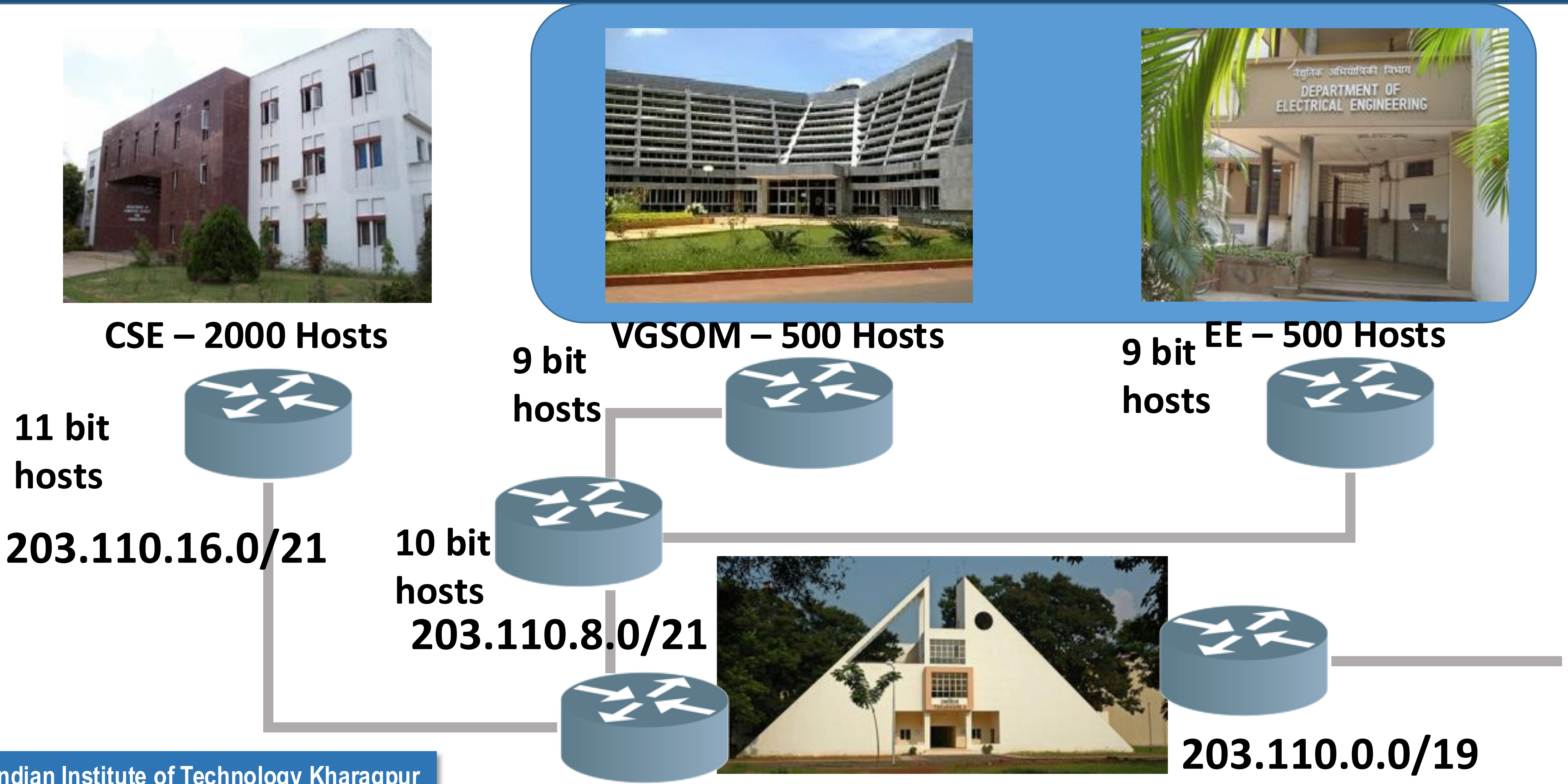


CIDR Example

CSE – 11 bits, VGSOM+EE – 10 bits

- Network address – 203.110.0.0/19, 203.110.000**XX**XXX.XXXXXXXX
- CSE network address 203.110.00010XXX.XXXXXXXX (203.110.16.0/21)
- VGSOM+EE network address 203.110.00001XXX.XXXXXXXX (203.110.8.0/21)

CIDR Example



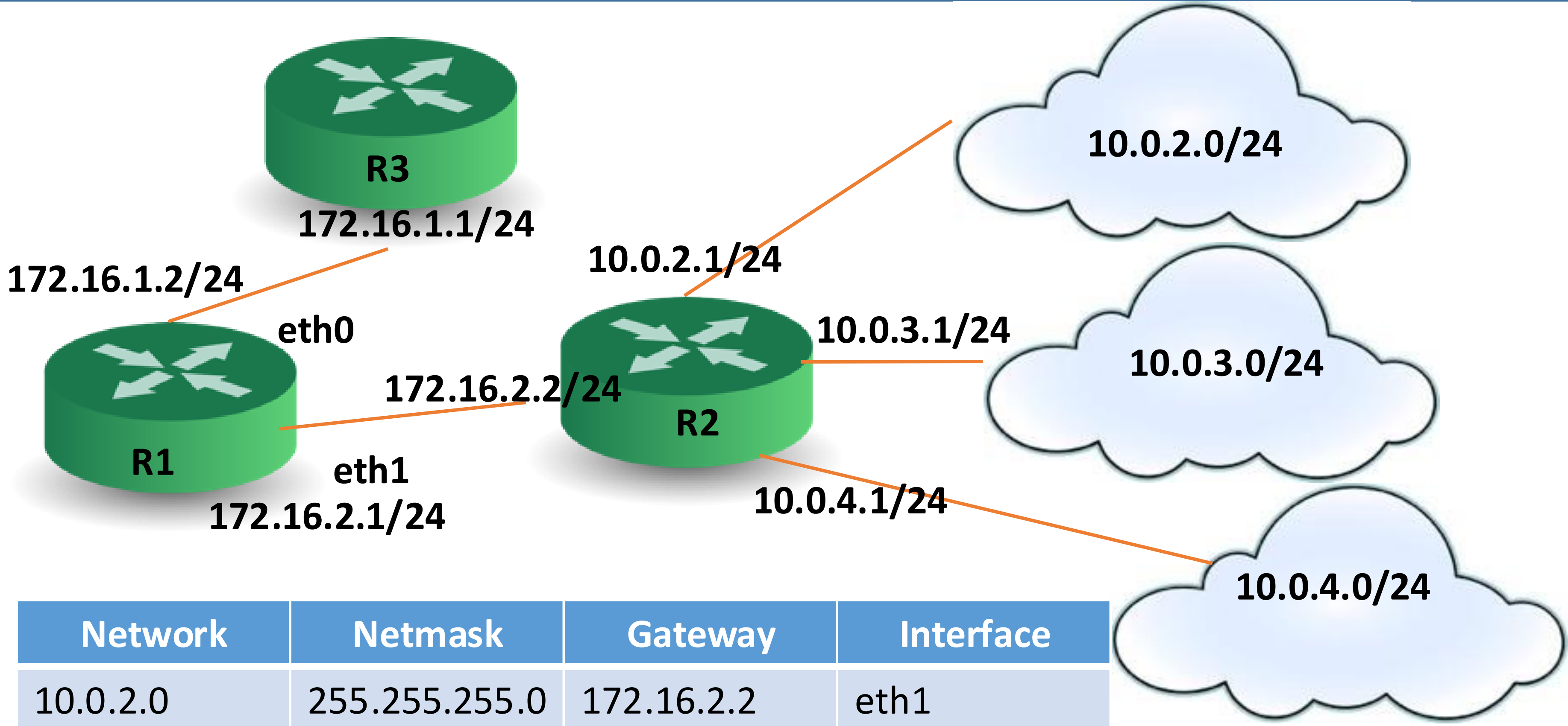
CIDR Example

VGSOM – 9 bits, EE – 9 bits

- Network address – 203.110.8.0/21, 203.110.00001XXX.XXXXXXXX
- VGSOM network address 203.110.0000110X.XXXXXXXX (203.110.12.0/23)
- EE network address 203.110.0000101X.XXXXXXXX (203.110.10.0/23)

CIDR Example

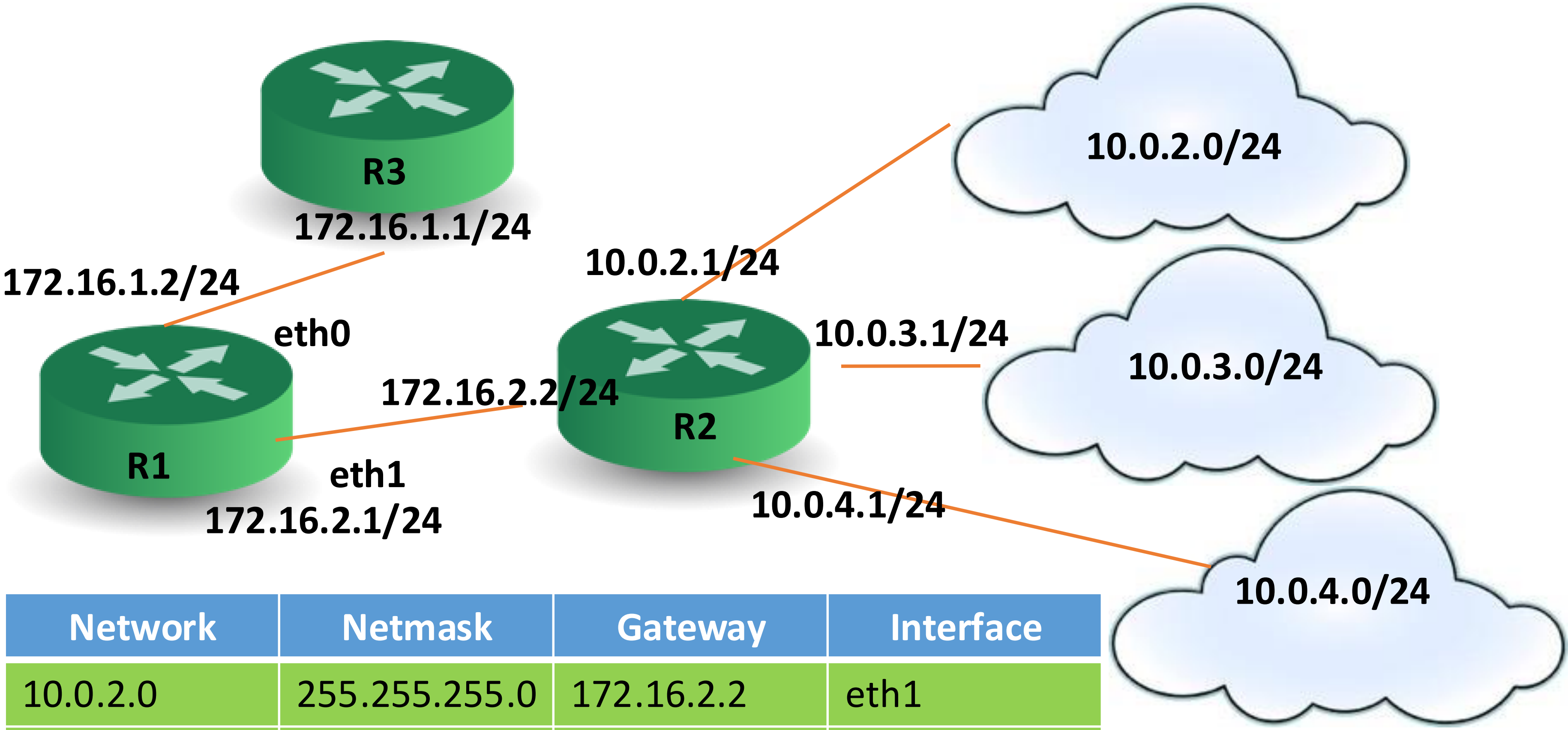
CIDR – Routing Table Construction



Routing Table for R1

Network	Netmask	Gateway	Interface
10.0.2.0	255.255.255.0	172.16.2.2	eth1
10.0.3.0	255.255.255.0	172.16.2.2	eth1
10.0.4.0	255.255.255.0	172.16.2.2	eth1
0.0.0.0	0.0.0.0	172.16.1.1	eth0
172.16.1.0	255.255.255.0	172.16.1.1	eth0
172.16.2.0	255.255.255.0	172.16.2.2	eth1

CIDR – Routing Table Construction



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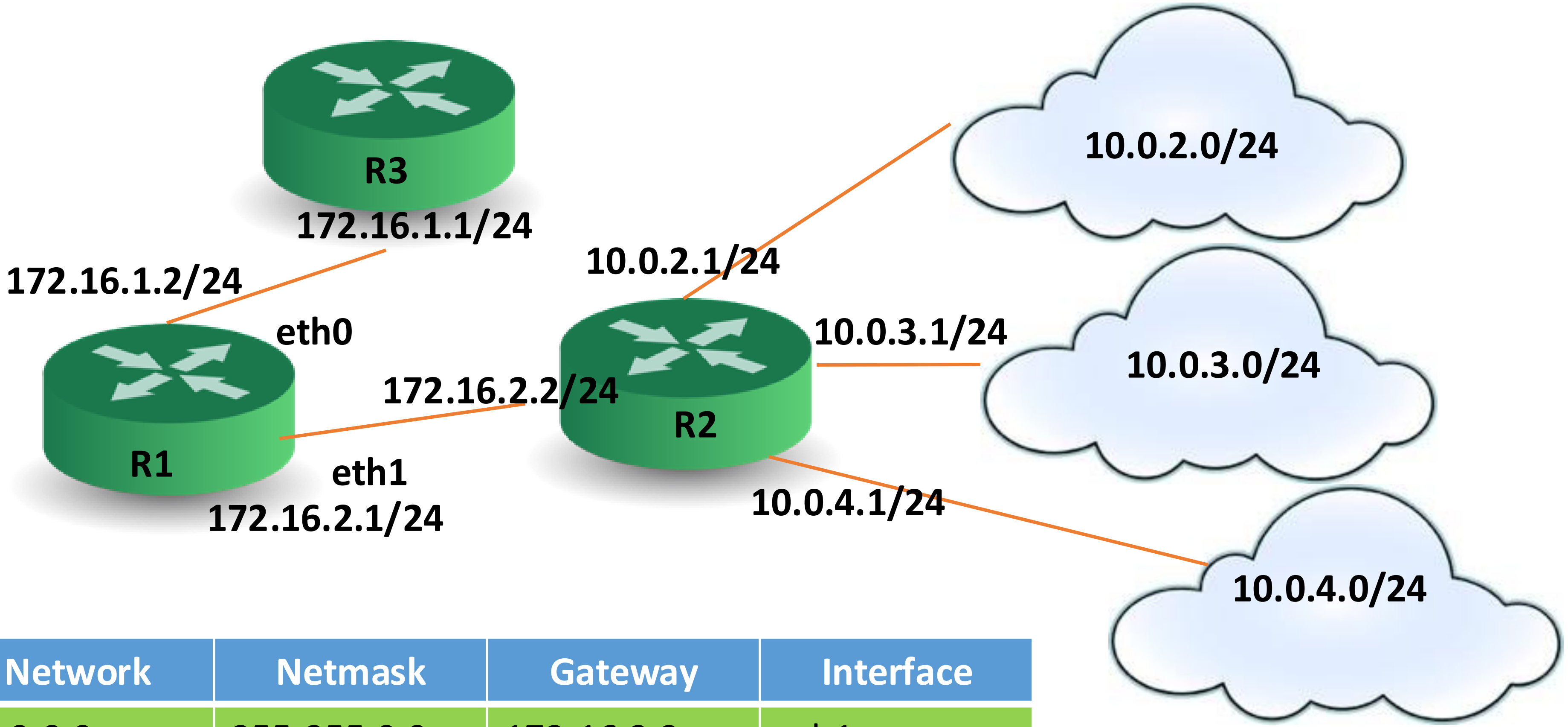
0.0.0.0	0.0.0.0	172.16.1.1	eth0
172.16.1.0	255.255.255.0	172.16.1.1	eth0
172.16.2.0	255.255.255.0	172.16.2.2	eth1

Compaction of routing table is possible

CIDR – Routing Table Construction



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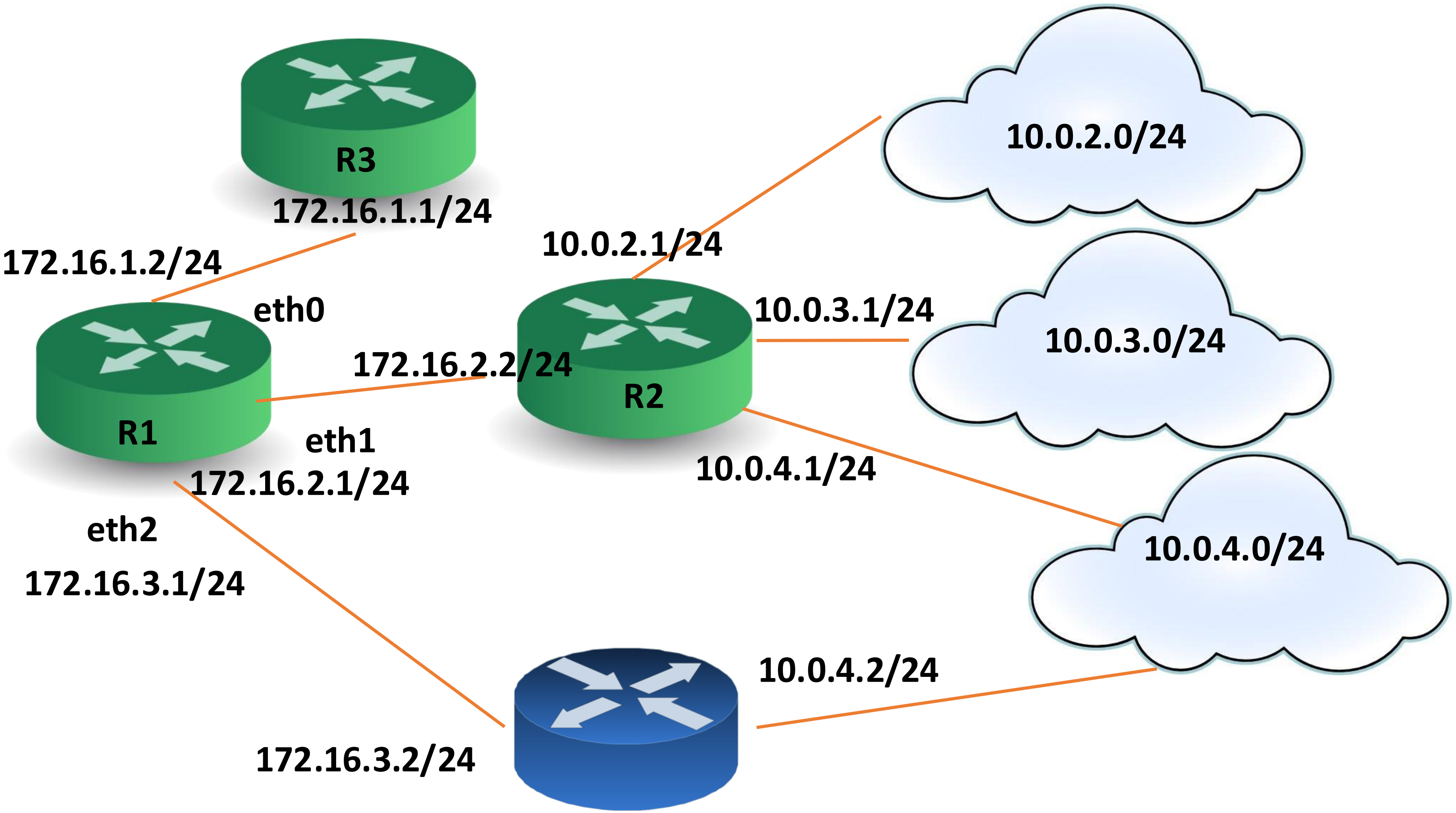
0.0.0.0	0.0.0.0	172.16.1.1	eth0
172.16.1.0	255.255.255.0	172.16.1.1	eth0
172.16.2.0	255.255.255.0	172.16.2.2	eth1

Compact Routing Table for R1

CIDR – Problem of Multihoming



CIDR – Longest Prefix Match

Supernetting is not always perfect !
There is always a possibility of duplicate entries

Network	Netmask	Gateway	Interface
10.0.0.0	255.255.0.0	172.16.2.2	eth1
10.0.4.0	255.255.255.0	172.16.3.2	eth2
0.0.0.0	0.0.0.0	172.16.1.1	eth0
172.16.1.0	255.255.255.0	172.16.1.1	eth0
172.16.2.0	255.255.255.0	172.16.2.2	eth1

Where to forward 10.0.4.8 ?

Longest prefix matching

longest prefix match

when looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** ********	eth0
11001000 00010111 00011000 ********	eth1
11001000 00010111 00011*** ********	eth2
otherwise	eth3

examples:

11001000 00010111 00010110 10100001 which interface?

11001000 00010111 00011000 10101010 which interface?

Source: Computer Networking: A Top-Down Approach (8th Ed) by Jim Kurose, Keith Ross

Longest prefix matching

longest prefix match

when looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range				Link interface
11001000	00010111	00010***	********	0
11001000	00010111	00011000	********	1
11001000	00010111	00011***	********	2
otherwise				3

match!

examples:

11001000 00010111 00010110 10100001 which interface?

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otherwise	3

match!

examples:

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11001000 00010111 00011*** ********	2
otherwise	3

match!

examples:

11001000 00010111 00010110 10100001 which interface?

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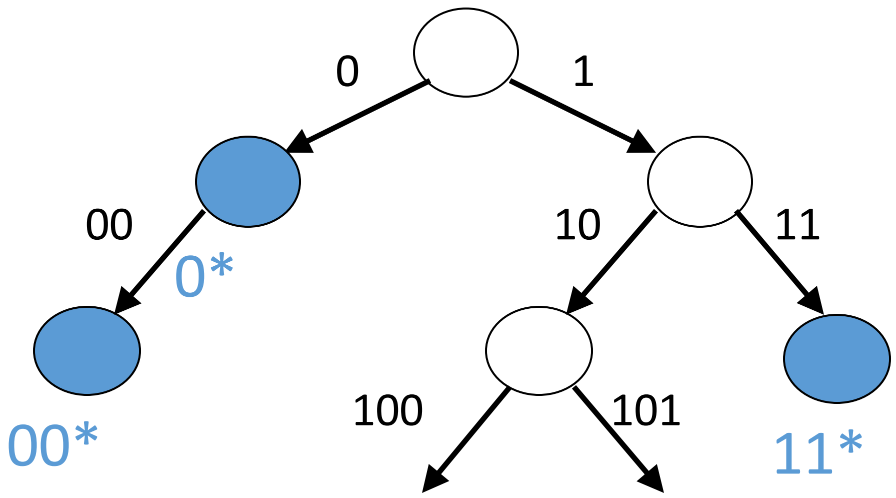
CIDR – Longest Prefix Match

- Use **Patricia Tree** (a compact representation of trie) for matching prefixes.



IP addresses: how to get one?

There are two questions:

1. How does a *host* get IP address within its network (host part of address)?
2. How does a *network* get IP address for itself (network part of address)

How does *host* get IP address?

- hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- DHCP: Dynamic Host Configuration Protocol: dynamically get address from as server
  - "plug-and-play"

DHCP: Dynamic Host Configuration Protocol

goal: host *dynamically* obtains IP address from network server when it "joins" network

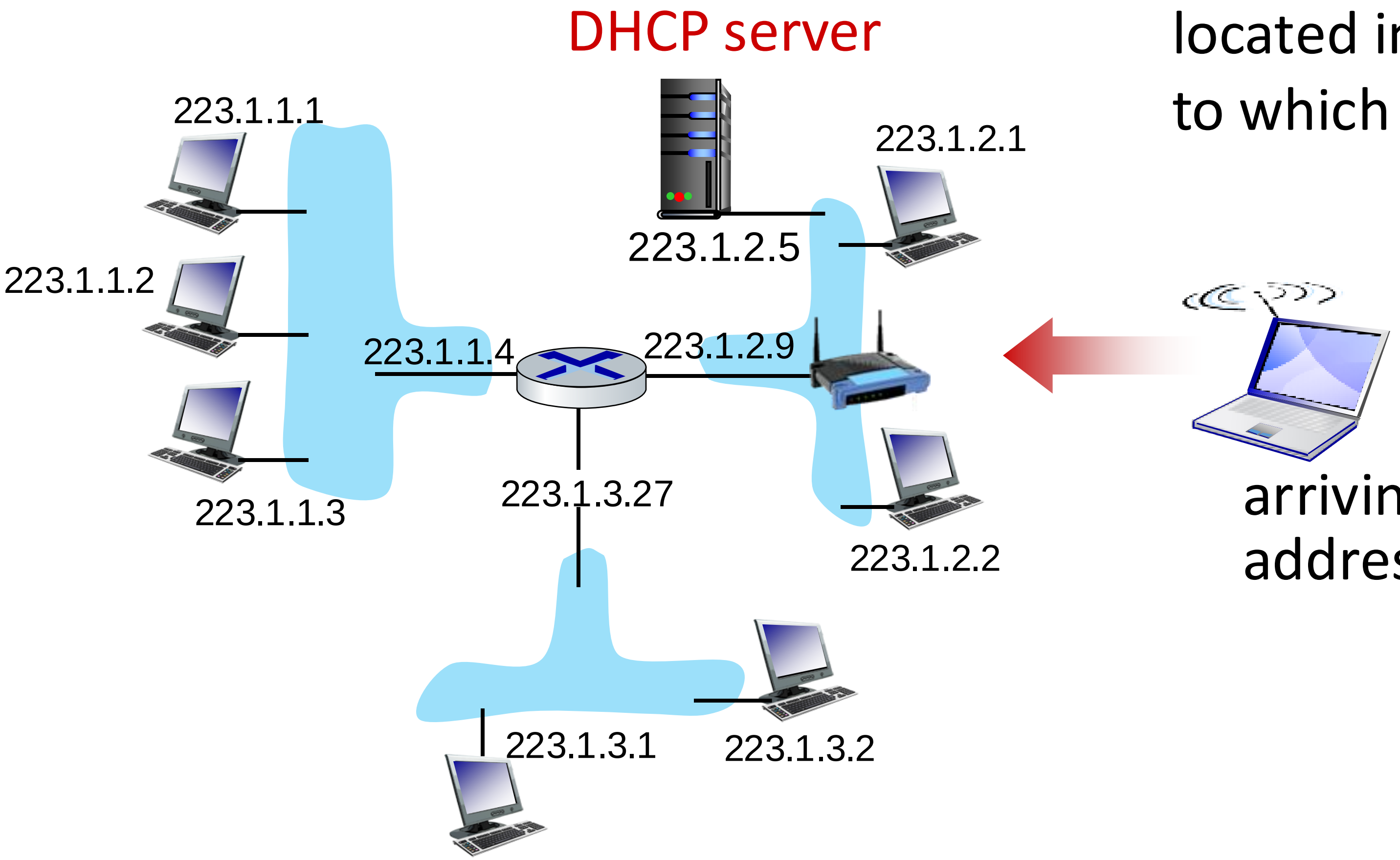
- can renew its lease on address in use
- allows reuse of addresses (only hold address while connected/on)
- support for mobile users who join/leave network

DHCP overview:

- host broadcasts DHCP discover msg [optional]
- DHCP server responds with DHCP offer msg [optional]
- host requests IP address: DHCP request msg
- DHCP server sends address: DHCP ack msg

Source: Computer Networking: A Top-Down Approach (8th Ed) by Jim Kurose, Keith Ross

DHCP client-server scenario

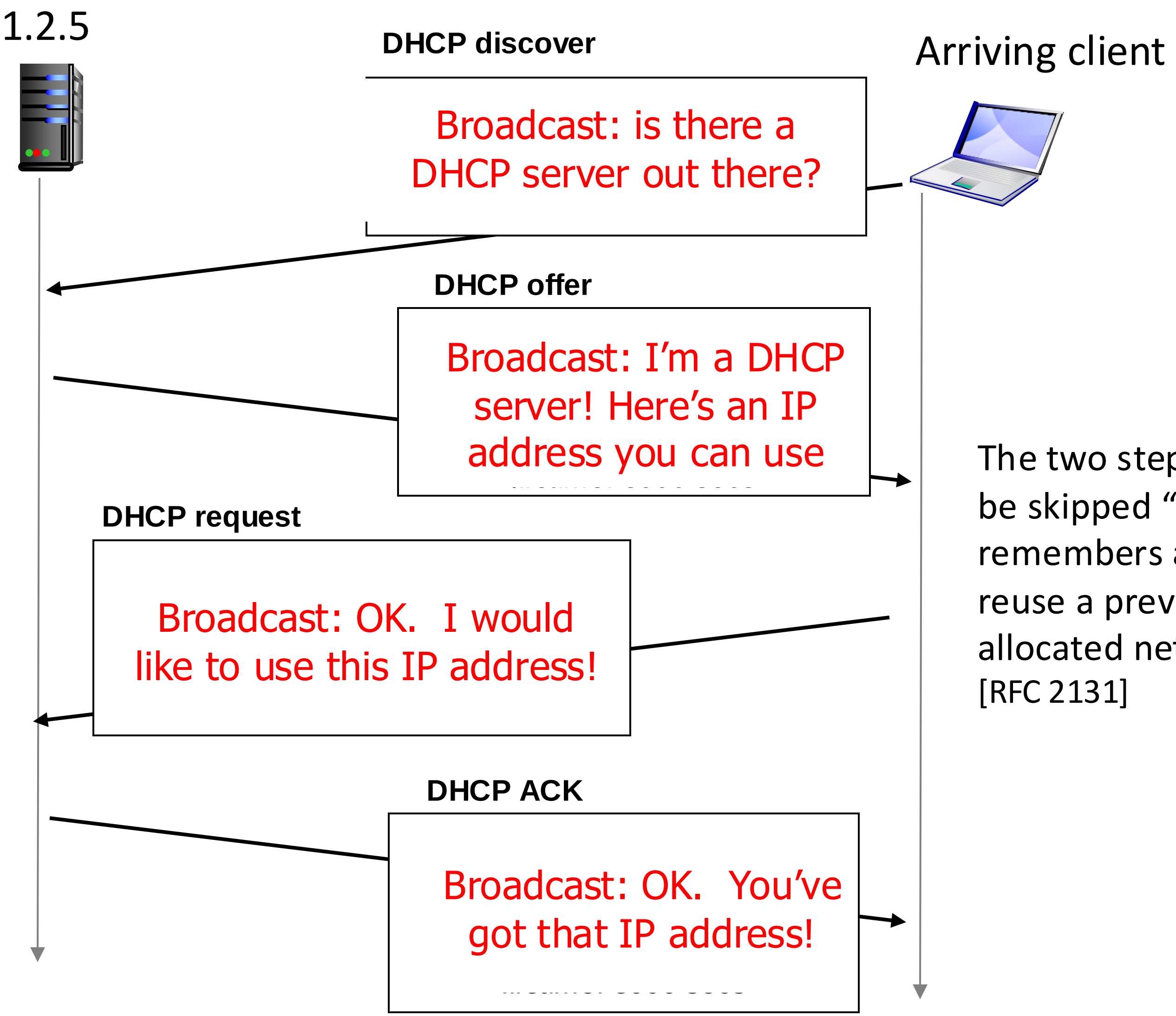


Typically, DHCP server will be co-located in router, serving all subnets to which router is attached

arriving DHCP client needs address in this network

Source: Computer Networking: A Top-Down Approach (8th Ed) by Jim Kurose, Keith Ross

DHCP client-server scenario



The two steps above can be skipped "if a client remembers and wishes to reuse a previously allocated network address" [RFC 2131]

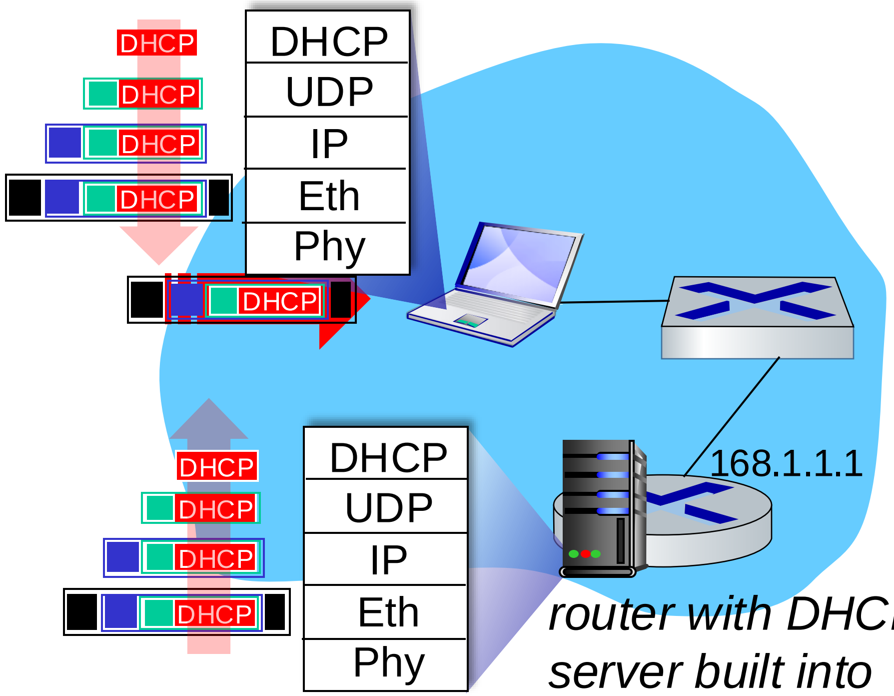
Source: Computer Networking: A Top-Down Approach (8^{th} Ed) by Jim Kurose, Keith Ross

DHCP: more than IP addresses

DHCP can return more than just allocated IP address on subnet:

- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating network versus host portion of address)

DHCP: example

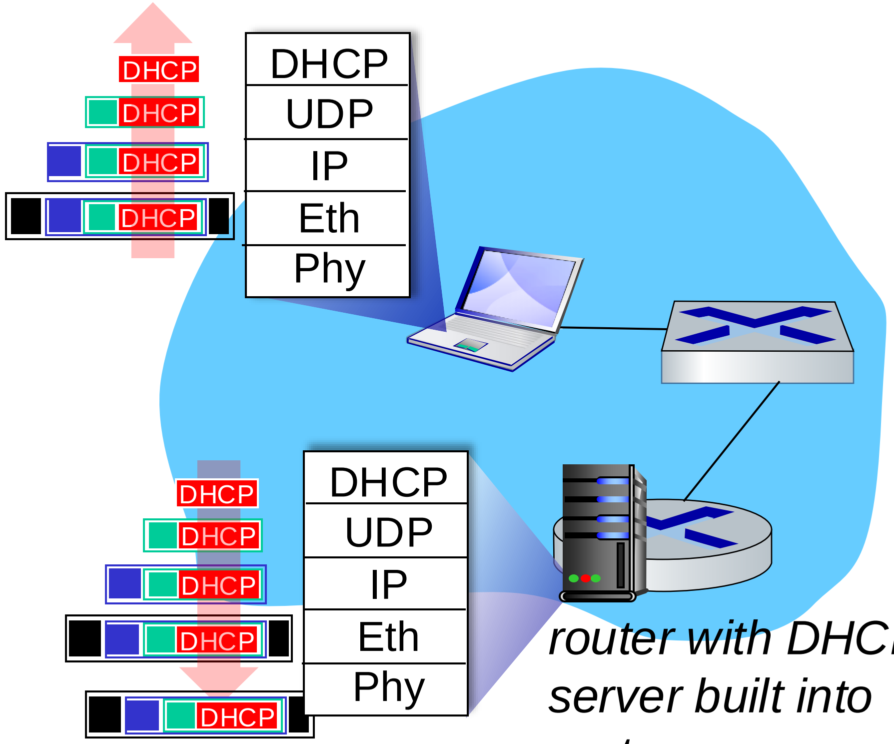


router with DHCP server built into router

- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FFFFFFFFFFFF) on LAN, received at router running DHCP server
- Ethernet demux'ed to IP demux'ed, UDP demux'ed to DHCP

Source: Computer Networking: A Top-Down Approach (8th Ed) by Jim Kurose, Keith Ross

DHCP: example



router with DHCP server built into router

- DCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulated DHCP server reply forwarded to client, demuxing up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

Source: Computer Networking: A Top-Down Approach (8th Ed) by Jim Kurose, Keith Ross

IP addressing: last words ...

Q: how does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers http://www.icann.org/

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , …) management

Q: are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

"Who the hell knew how much address space we needed?" Vint Cerf (reflecting on decision to make IPv4 address 32 bits long)

Next, we’ll look how to construct the routing table for the Internet …